

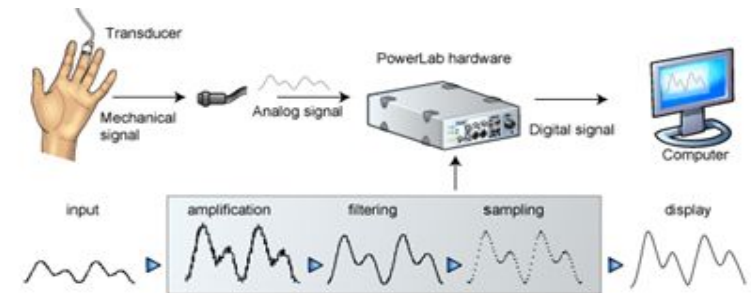
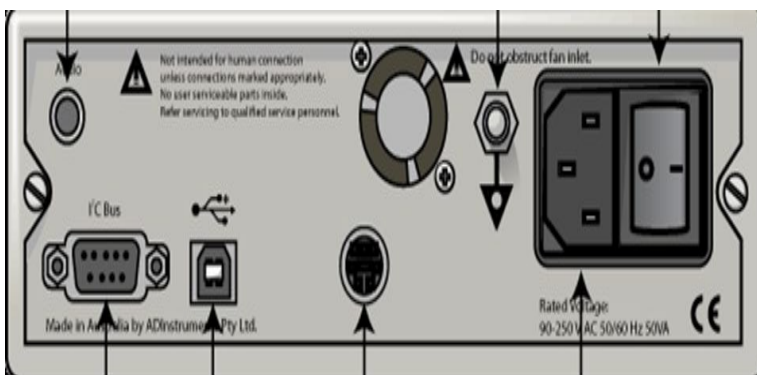


INTRODUCTION TO LABTUTOR (POWERLAB)

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Dr Eshratul Ebad Institute of Oral Health
Sciences

Dow University of Health Sciences



Data acquisition with a PowerLab system

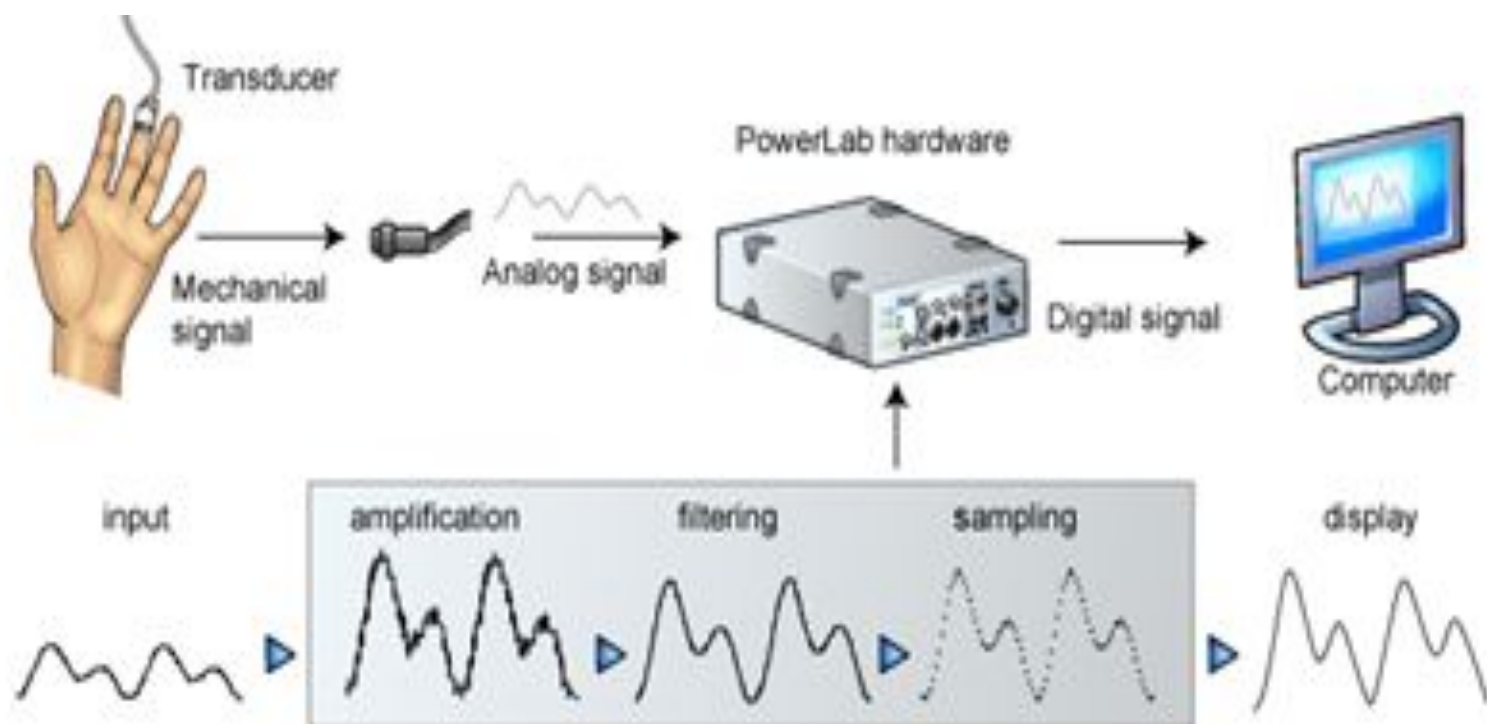
INTRODUCTION TO POWERLAB:

Definition:

A system consisting of hardware which is intended to acquire data and transduce one type of signal into another

e.g.,

- Mechanical  Digital signals
- Electrical  Digital signals
- Pressure  Digital signals



Data acquisition with a PowerLab system

INTRODUCTION TO LABTUTOR

- An HTML-based **software** package
- Laboratory teaching
- Used in conjunction with AD Instruments' Power Lab
- Lab Tutor is used in **sampling, digitizing** and **storage** of experimental data
- Incorporates experimental protocols with **data acquisition** (the act of obtaining or beginning to have something, or something obtained), **analysis** and **report writing**

INTRODUCTION TO LABTUTOR

- Usage: Physiology, Pharmacology, Biomedical engineering, Sports/exercise studies and Psychophysiology laboratories
- Record and analyze physiological signals from human or animal subjects

ADINSTRUMENTS
PowerLab 4/25T



Power



Status



—



Trigger



Output



1



2



Input

Isolated Stimulator

Status



Green = Pulse
Yellow = OFF



WARNING
100 Volts @ 30mA

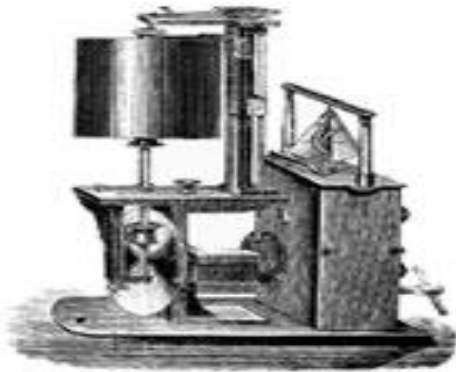


Bio Amp
(Input 3 & 4)



EVOLUTION:

1890s



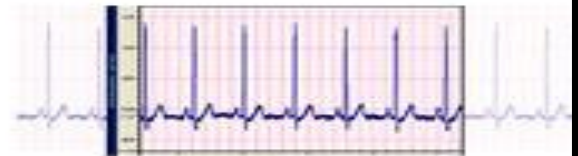
**Drum kymograph
according to Ludwig**

1960s



Grass Polygraph

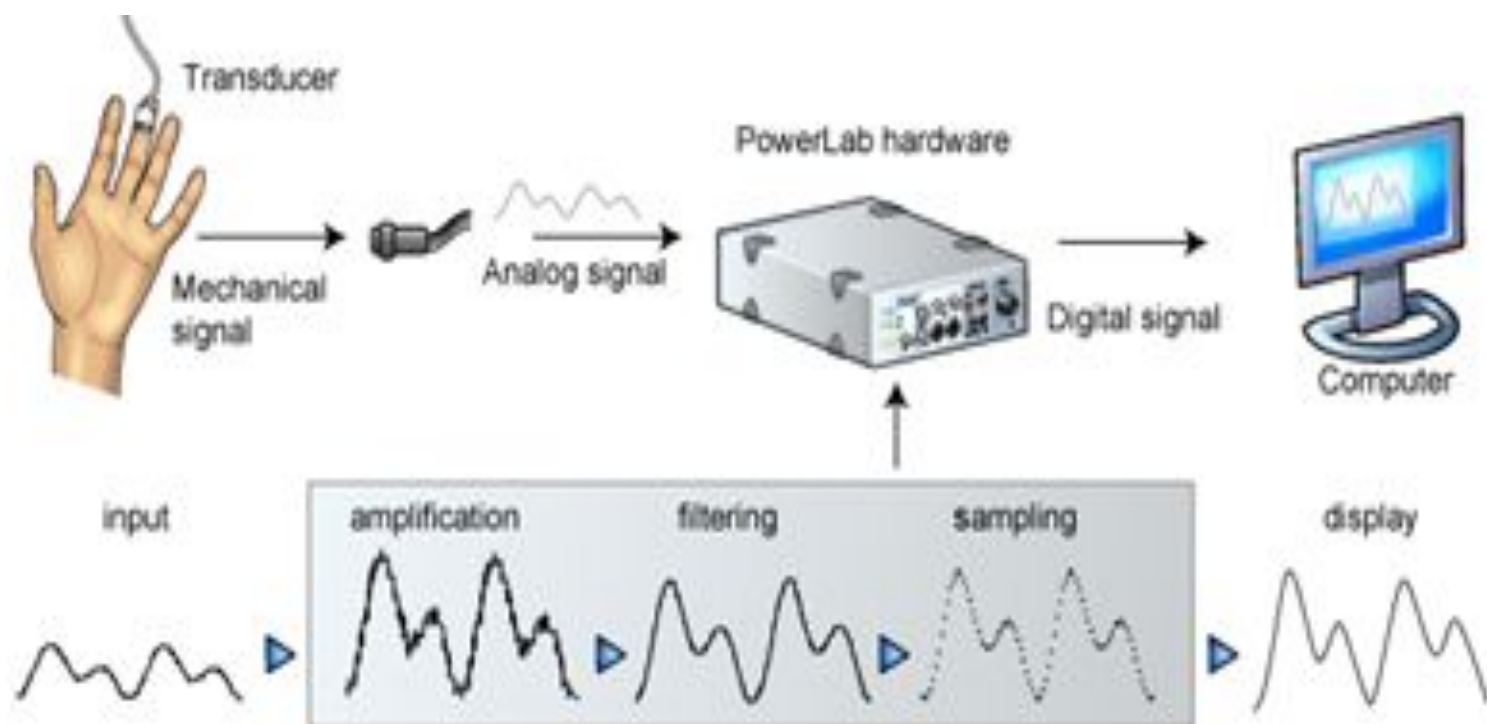
Today



**The PowerLab and a
LabTutor panel,
displaying a section
of recorded data**

HOW IS DATA ACQUIRED?

- External signals detected are **converted** into analog electrical signal
- Signals are **amplified** to amplify signals and **filtered** to remove unwanted frequencies or noise
- Analog signal is **multiplexed** from an analog to digital converter
- The digitized signal is **transmitted** to the computer using USB connection
- Software **receives, displays, analyses** and **records** data.



Data acquisition with a PowerLab system

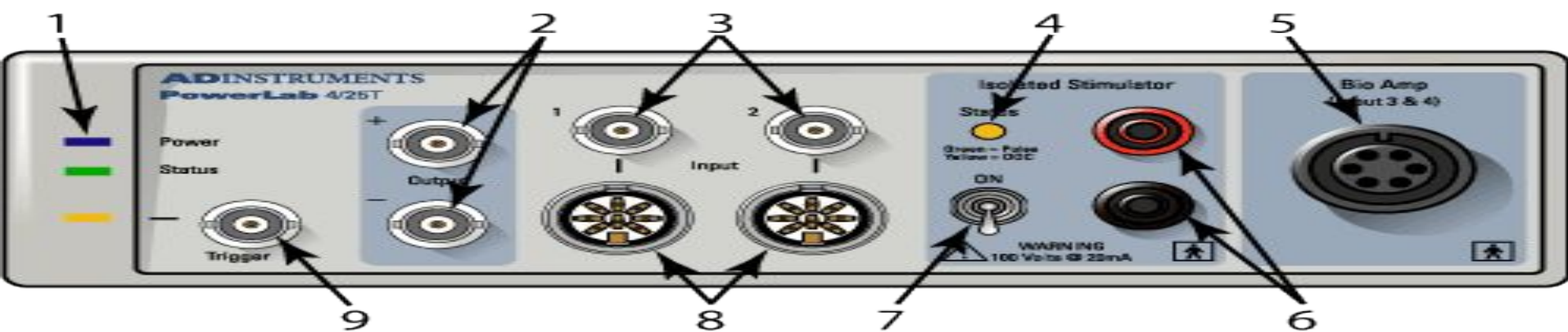
TRANSDUCERS:

- Transducers are the leads that convert the one type of signals(Analog) in to the digital signals.
- Transducers are connected to the power lab and to the human subject.

TRANSDUCERS:

- Nerve and muscle stimulation  Nerve and muscle Practicals
 - Pressure transducer  Muscle Practicals
 - ECG Leads and Electrodes  ECG Practical
 - Pulse Transducer  Pulse Practical
 - Pneumotacometer  Respiratory Practicals
- These transducers are connected to the powerlab on the respective ports present on front of powerlab.





1. Power indicator light

2. Analog output connections: NOT safe for direct connection to humans

3. Analog inputs

4. Isolated Stimulator status light

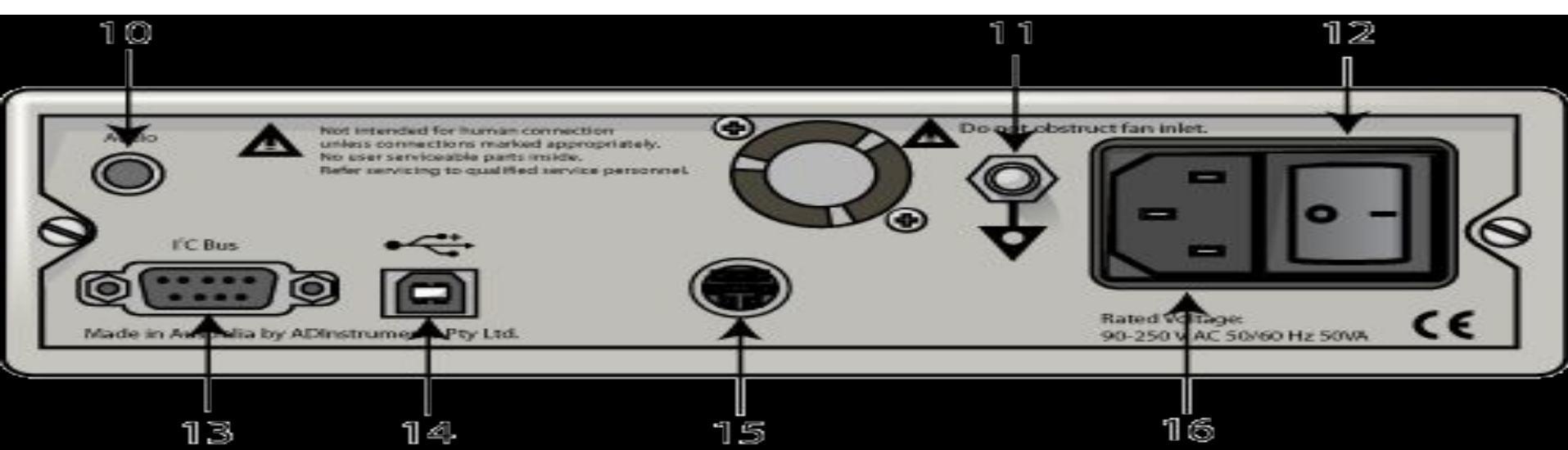
5. Dual Bio Amp input

6. Isolated Stimulator outputs

7. Isolated Stimulator switch

8. Pod ports

9. Trigger input



10. Audio output connector

11. Earthing post

12. Power switch

13. I2C connector

14. USB connector

15. Serial Port connector

16. Power cord connector

SIGNAL TRANSDUCERS

Virtually any transducer that generates an analog voltage between $\pm 10\text{V}$ may be attached to the Power Lab, either directly or through one of the signal conditioners mentioned above.

EVENT MARKERS

Some transducers may be used to mark an event marker, like this pushbutton switch.



STIMULATION OF NERVES AND MUSCLES

- Power Lab also has the ability to produce an output voltage to induce a response in a nerve or muscle
- This stimulating bar electrode connects to the Isolated Stimulator on the Power Lab.



FORCE OR PRESSURE TRANSDUCERS

Some transducers measure force like this hand dynamometer (force transducer) used in muscle fatigue practicals.



CHEMICAL PROPERTIES

Chemical properties may also be measured such as conductivity, dissolved oxygen, and pH.



MOVE TO END BUTTON



HORIZONTAL COMPRESSION BUTTONS



VERTICAL SCALING BUTTONS



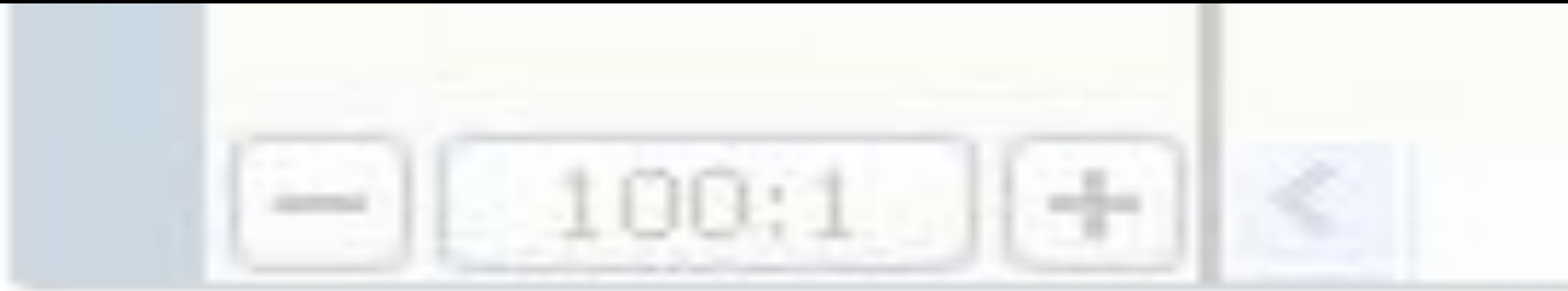
AUTOSCALE BUTTONS



DEFAULT SCALE BUTTON



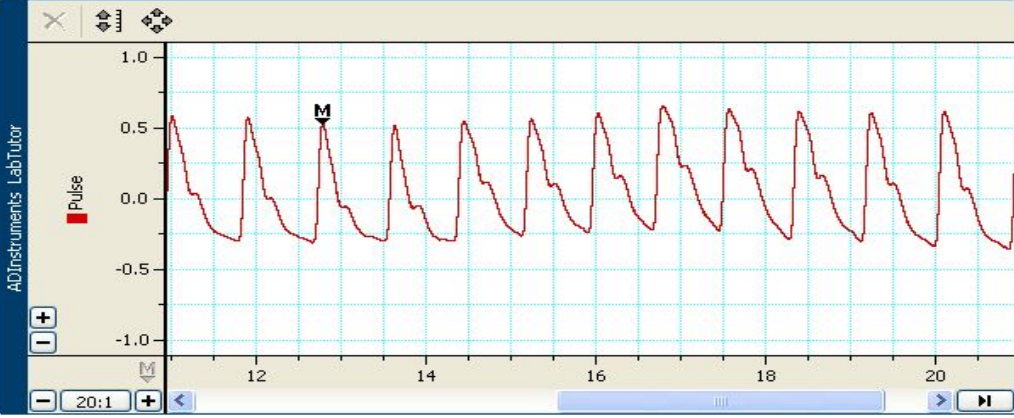
COMMENT PANEL



MAKING MEASUREMENTS:

**LabTutor®**
Introduction to LabTutor

Analysis - Tables 11 / 15 Previous Next



ADInstruments LabTutor

Pulse

Time: 20:1

The graph displays a red pulse waveform on a grid. The y-axis ranges from -1.0 to 1.0, and the x-axis ranges from 12 to 20. A cursor is positioned over a peak, labeled 'M'.

Making Measurements

You can make measurements and add the relevant numbers into a Table.

1. Place the cursor over the data at the desired point and click to place the selected data in the **Value** panel.
2. To insert a value into a table, drag it from the Value panel into the appropriate cell of the table.
3. You can also type information, including values, into any cell in the Table.
4. Complete the Table using the data you have already recorded, by clicking a peak of the pulse recording and transferring the time and amplitude values into the first row of the table. Repeat for the next three peaks of the pulse.

Time

Value

Pulse

Value

Pulse Amplitude

Table

Time	Pulse Amplitude

background

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ADINSTRUMENTS

Done

Internet

100%

MARKER

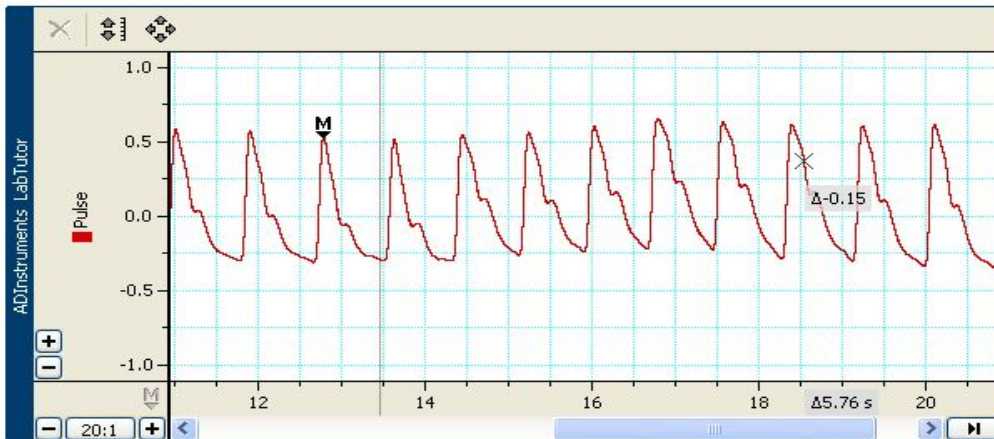


LabTutor®

Introduction to LabTutor

Analysis - The Marker

12 / 15 Previous Next



The Marker

When the Marker is in use, values and times displayed in the Value panel are relative to the Marker position. When not in use, the Marker resides in a dock at the bottom left of the LabTutor panel; from here it can be dragged and dropped on to any part of the data. To return the Marker to its dock, simply click in the dock area.

1. Drag the Marker from its home and drop it somewhere on the trace. The Marker does not have to be placed exactly on the waveform. When released, the Marker will drop and attach itself to the waveform.

Values placed in a Value panel will now be prefaced by a Δ (delta) symbol, to indicate that the measurement is relative to the Marker's position.

2. Complete the Table using the Marker. Place the Marker on a peak of the pulse recording. Click on the following trough to add the values for time and amplitude to their respective Value panels. Drag these values into the table and repeat for the next three peaks of the pulse (see [example](#)).
3. Remove the Marker from the waveform by clicking in the Marker dock.

Value
Time
 $\Delta 0.68$ s

Value
Pulse
 $\Delta 0.81$

Table	Δ Pulse Amplitude	
	Δ Time	Δ Pulse Amplitude
	0.86	0.01
	6.44	0.04

background

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ADINSTRUMENTS

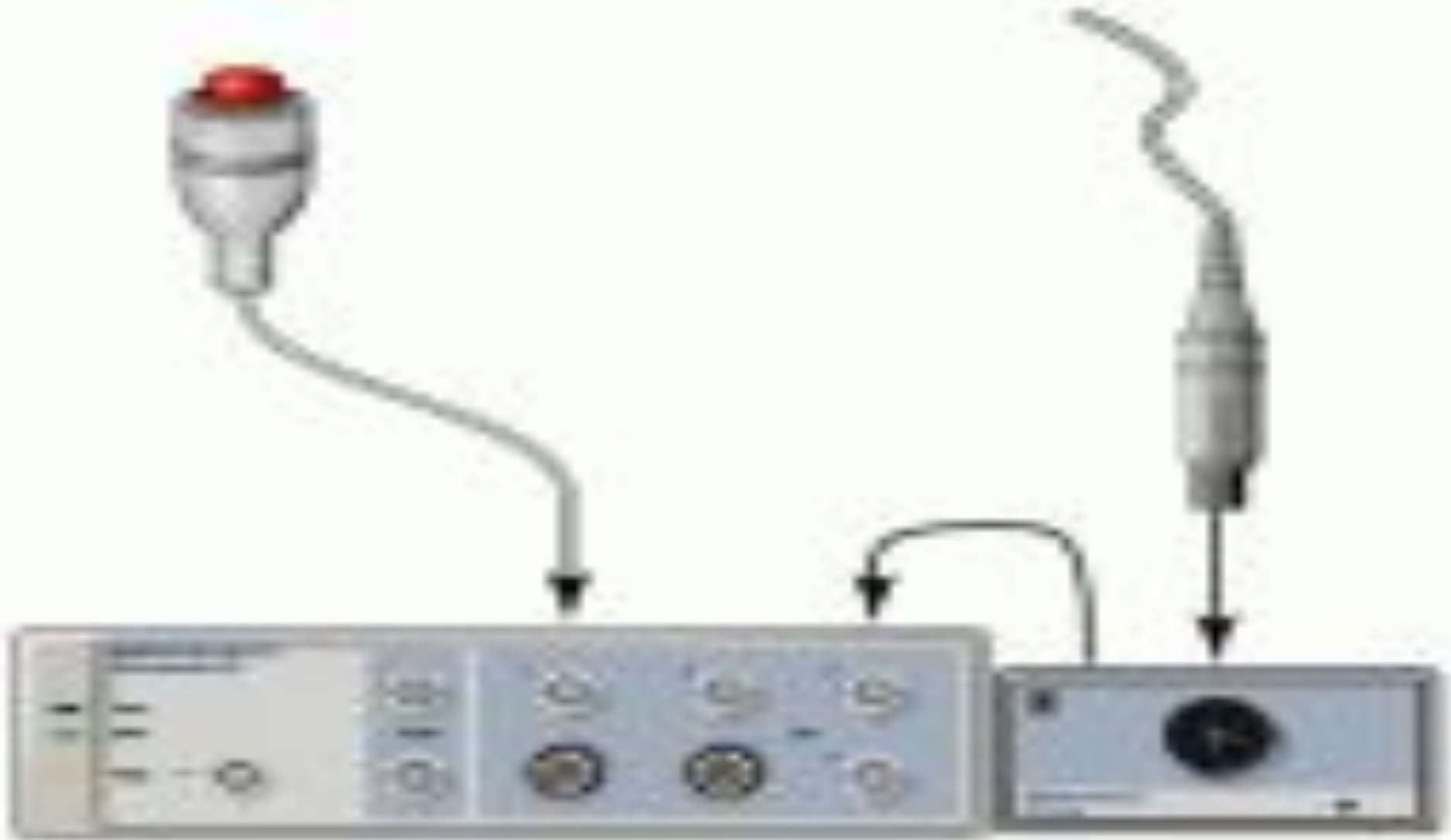
CLEAR SELECTION



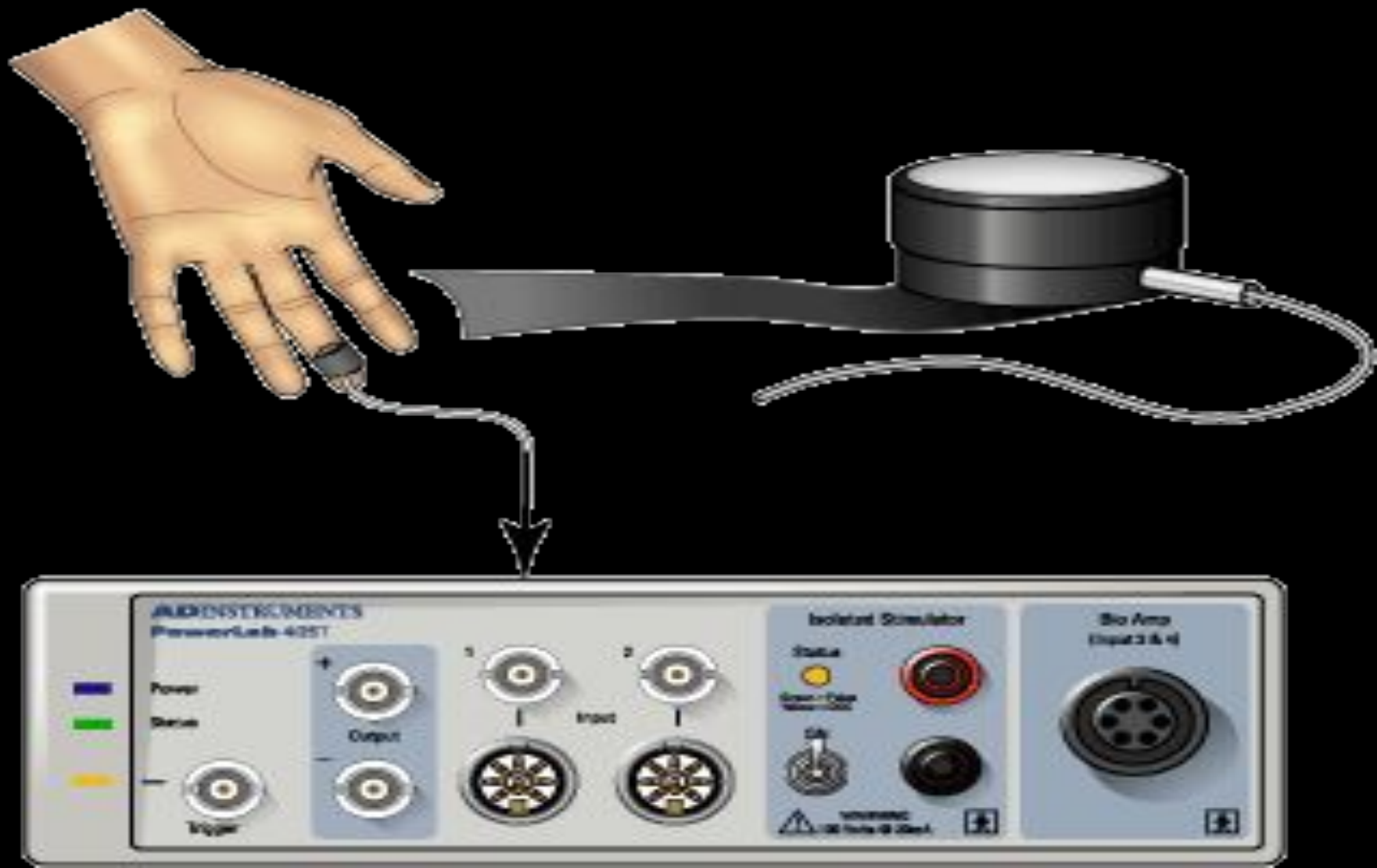
RETORT STAND MICROPOSITIONER MUSCLE HOLDER



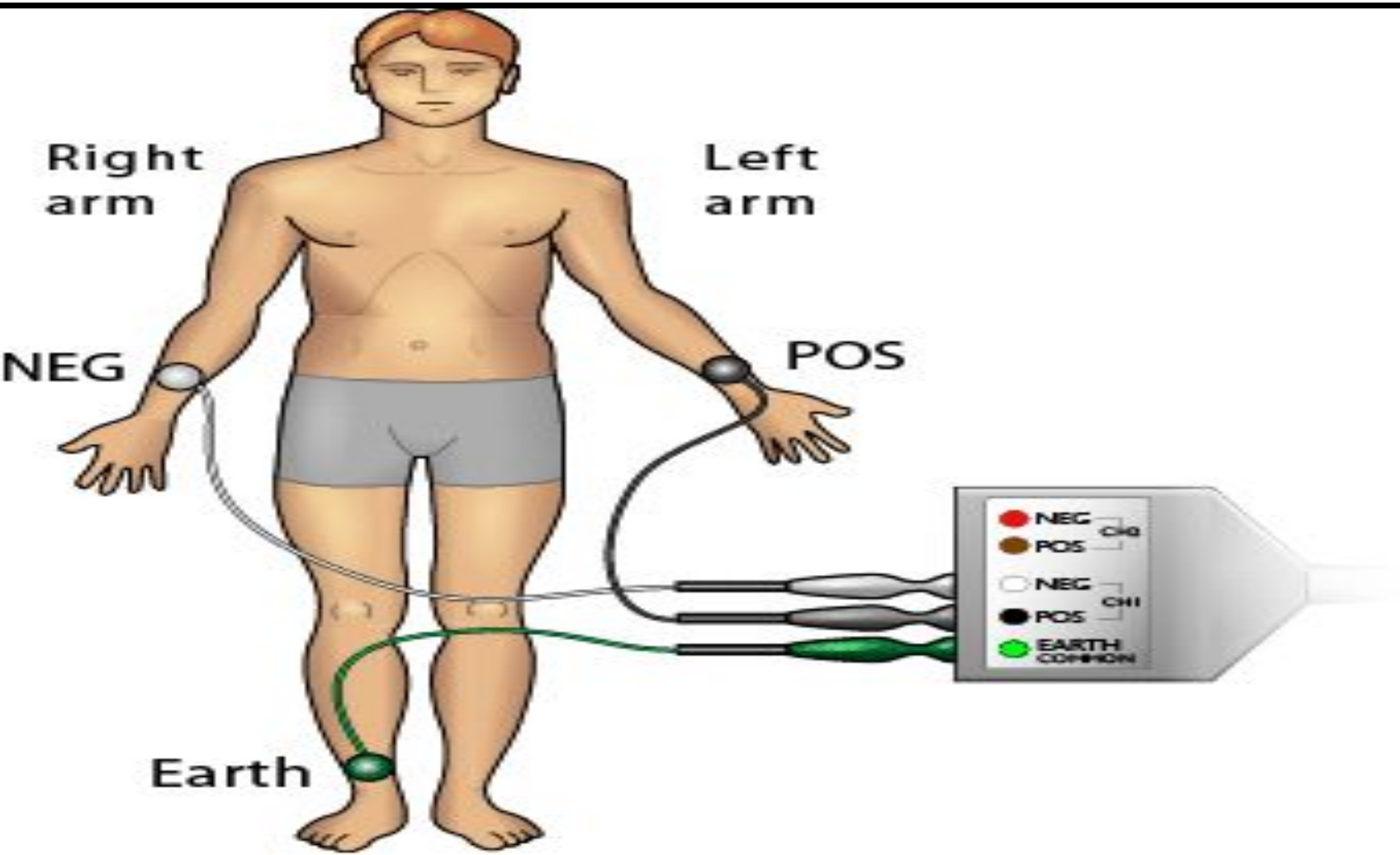
PUSH-BUTTON SWITCH



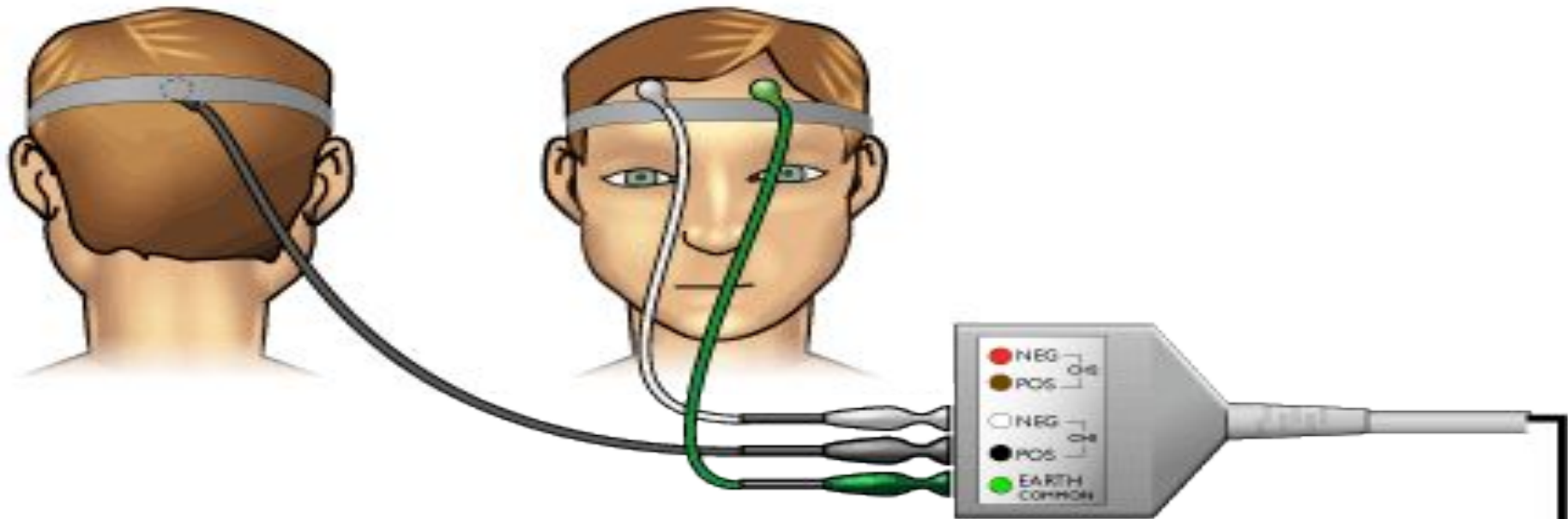
PULSE TRANSDUCER



ECG ELECTRODES



ELECTROENCEPHALOGRAMS (EEGs)



RESPIRATORY BELT

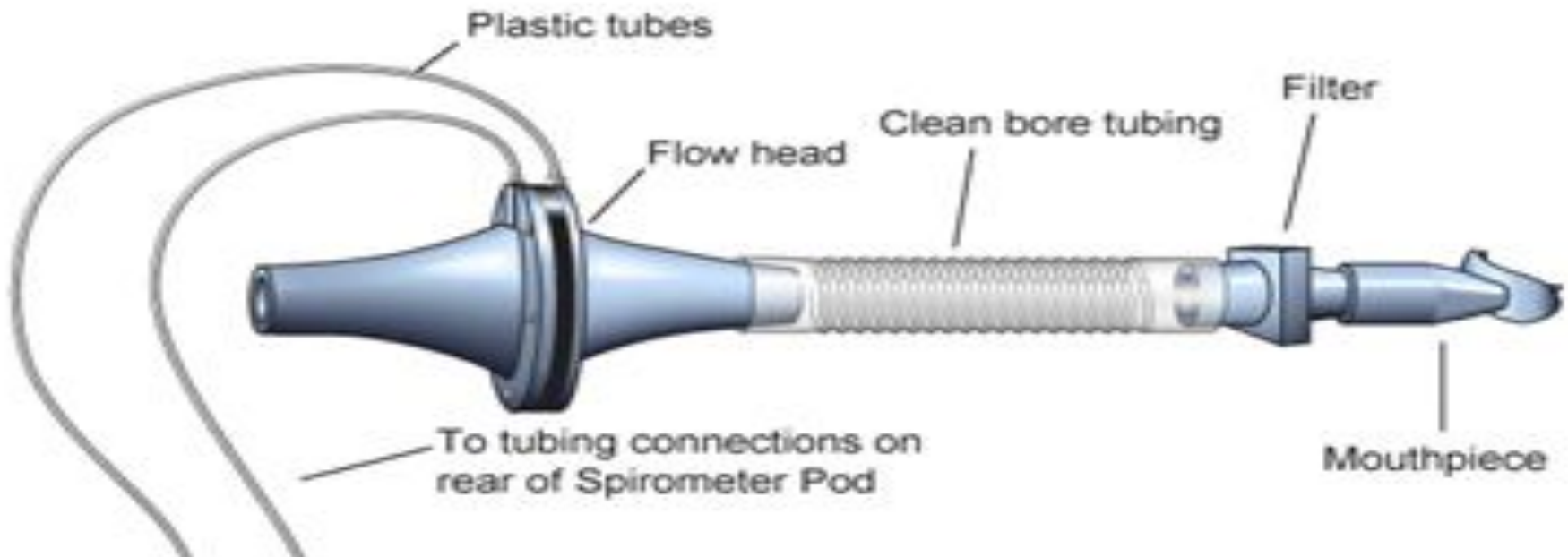


RESPIRATORY BELT & FINGER PULSE TRANSDUCER



PNEUMOTACOMETER

- One end of the Pneumotacometer is connected with the volunteer and other end is connected to the Power lab.



THANK YOU